

DATA MIND 3.0

Provisional Module-and-Interface Freeze

Governing rule. No module gains mathematical authority merely because it is intelligent, predictive, reflective, highly cleared, or favored by the controller.

The Verifier remains the sole mathematical trust anchor. BANK contains verifier-accepted material; FUTUREBANK contains represented possibilities; all other intelligence changes search, allocation, visibility, or presentation rather than theoremhood.

Core architectural distinction

Verifier: decides whether a mathematical certificate is accepted. **BANK:** preserves verified reusable mathematics. **FUTUREBANK:** quarantines speculative futures. **Professor:** allocates resources. **Partial Credit:** measures unfinished progress. **Sentinel:** enforces runtime safety. **Awareness/Clearance:** controls what a module may formally know or inspect.

DATA MIND 3.0 interface matrix

Entity	Main job	Reads	Produces / writes	BANK / trust boundary
Verifier (VE)	Mathematical trust anchor	Candidate certificates; admissible dependencies	ACCEPT / REJECT; verifier record	Sole admission gate to trusted mathematics
P1/P2	Proof settlement	Active state; Librarian shelf; Compass; partner advice	Proof steps; candidate lemmas/certificates	Read; deposits only through VE
R1/R2	Refutation settlement	Same typed views	Refutation candidates/certificates	Read; deposits through VE
I1/I2	Independence settlement	Same typed views	Independence candidates/certificates	Read; deposits through VE
C1/C2	Integrity / contradiction settlement	Same typed views	Integrity/contradiction candidates	Read; deposits through VE
BANK	Verified past	Verifier-accepted objects	Append-only verified records	Protected, typed, provenance-aware memory
FUTUREBANK	Possible futures	Speculative branches	Hypotheses; histories; dependency graphs	Never trusted merely by membership
Librarian	Retrieve relevant verified knowledge	Clearance-permitted BANK	Active working shelf	Heavy read; never changes theoremhood
Shortcut Engine	Cheap verified macro transitions	BANK / Librarian shelf	Macro proposals; expandable certificates	Macro admission remains verifier-gated
Presentation Manager / Trading	Certified equivalent presentations	Presentation; verified trades; cost model	Candidate P'; translation/equivalence certificates	Presentation authority, not truth authority
Quotient Hunter	Quotients, invariants, obstructions	Presentation; BANK; search state	Quotient/invariant proposals	Hard pruning only after certification
Proof Compass K1	Rank mathematical moves	Proof state; features; permitted BANK evidence	Transition scores	Advice only
Policy Compass K2	Rank controller actions	Global control state	Policy-action scores	Advice only
Partial Credit	Measure unfinished progress	State; horizons; verified structure; trajectories	PC vector; uncertainty; trajectory	Measurement, never certification
Professor	Allocate resources	PC; Compass; tensor; Scout; Sentinel risk; BANK stats	Allocation plan	Scheduling authority only
Scout	Cheap bounded probes	Proposed action; sandbox budget	Empirical probe report	Strictly budget-limited
Quicksand Monitor	Detect stagnant attractive basins	PC/horizon/certificate-density/cost	Stagnation or bailout signal	Diagnostic only
Skeptic / Counselor	Challenge misleading estimates	PC calibration; Professor predictions; Compass; history	Challenge/calibration report	Independent critic, no proof authority
Creativity Engine	Generate structurally different ideas	Search summaries; failures	Novel candidate strategies	Speculative output
Revision Engine	Change search controls	Quicksand; Skeptic; Professor state	Revised policy/control coordinates	Strategy authority only
Sentinel	Operational containment/security	RAM/time/growth/access/worker behavior	Allow; throttle; quarantine; kill; checkpoint	Safety veto; never changes theoremhood
Awareness / Self-Model	Clearance and formal system model	Authorized telemetry/state descriptions	Clearance-compatible views	Controls knowledge/visibility, not truth

Entity	Main job	Reads	Produces / writes	BANK / trust boundary
Historian / Transaction Ledger	Immutable episode history + analysis	Decisions; resources; PC; trades; BANK hits; Sentinel events	Event ledger; episode traces	Audit record, not mathematical memory
Distiller	Extract reusable structure after episodes	Proof; history; trades; QH; BANK hits	Lemma/shortcut/invariant/training proposals	Mathematical proposals still pass VE
Learner	Train horizon/gain/PC/retrieval/policy models	Target-clean historical data	Frozen model versions	Heuristic learning never rewrites BANK truth
Baseline Lane	Protected untouched search	Original presentation	Ordinary candidate certificates	Protected resource share
Benchmark / Evaluation	Ablations and falsifiability	Logs; certificates; costs	Metrics/reports	No trusted-state mutation
Reflective Orchestrator	Run iterative cycle	Module outputs	Calls; events; coordination	No special trust merely by orchestration

The wiring backbone

Three principal information paths are frozen before policy implementation.

1. Mathematical path

Search / QH / Trading / Distiller -> candidate -> Verifier -> BANK

There is no other path into BANK. A learned, speculative, high-partial-credit, or high-clearance claim remains outside trusted mathematical memory until verified.

2. Fast adaptive loop

Search -> Partial Credit -> Professor -> {Scout, allocation, Revision} -> Search

Professor does not accept Partial Credit blindly. Skeptic challenges calibration, Quicksand supplies stagnation evidence, Sentinel supplies operational risk, Compass supplies directional estimates, and Scout can cheaply probe an action before a major allocation.

3. Slow learning loop

Episode -> Historian -> Distiller / Learner -> {Verifier -> BANK | frozen learned models} -> future episodes

This separates two forms of learning: reusable mathematics must be verified before entering BANK, while heuristic lessons are stored as learned policy/retrieval models and remain non-authoritative.

Trading has a specific wire

Trading does not sit in the middle of every inference. The Presentation Manager proposes a bridge from one deductive presentation to another; certification establishes admissibility; then ordinary search, QH, BANK macros, Compass, Professor, Scout, and other search machinery may operate within the selected presentation.

$P_0 \leftrightarrow P_1 \leftrightarrow P_2 \dots$ (certified proof translations / equivalence bridges)

The protected original presentation remains available as a trust-anchor/baseline route. Trading changes search geometry; it does not itself assert speed or mathematical truth.

BANK = vertical reuse; Trading = horizontal movement.

BANK lets DATA MIND reuse verified knowledge through time and across agents. Trading lets DATA MIND move sideways among certified presentations of the same mathematics.

Clearance and separation are wired from day one

DATA MIND 3.0 should not begin as one giant object with unrestricted shared state. Each module receives an explicit interface and a permitted view. A generic BANK view can be written schematically as:

$BANKView(M) = \{ b \text{ in } BANK : classification(b) \leq clearance(M) \}$

Clearance determines what a module may know or touch; capability determines which operations it may execute; Sentinel determines whether the authorized behavior may safely continue. High clearance therefore does not imply authority to certify or mutate BANK.

Examples: Professor receives reports rather than Sentinel internals; P receives a Librarian shelf rather than unrestricted BANK internals; Scout receives a bounded sandbox; Creativity/Dreamer-style speculation writes to FUTUREBANK; Distiller may propose a reusable lemma or shortcut but cannot admit it to BANK.

New 3.0 additions

Librarian. BANK remembers everything verified; Librarian retrieves the small target-, role-, presentation-, and state-specific working shelf that is worth considering now.

Scout. Before Professor commits a large budget to a transformation or basin, Scout can run a cheap bounded probe and report partial-credit change, horizon movement, branching, BANK hits, RAM slope, novelty, and related evidence.

Historian. The old append-only Transaction Log becomes the factual substrate of an analytical Historian that records what was tried, why, at what resource cost, with what intermediate signals, and with what final result.

Skeptic / Counselor. An independent critic tests whether Partial Credit, Compass values, Professor predictions, and presentation-gain estimates remain calibrated instead of becoming self-reinforcing proxies.

Distiller. After a verified success or informative failure, Distiller asks what reusable structure should survive: a lemma, macro shortcut, invariant, useful trade, proof motif, or training example.

Security and operational protection

Separation is a security property, not merely a software-engineering preference. No module receives another module's internal mutable state merely because both belong to DATA MIND. Communication crosses explicit authorized interfaces.

Sentinel is independent of proof authority. A worker can be fully authorized to process a theorem and still be throttled or quarantined for abnormal RAM, time, process growth, queue growth, verifier-failure streaks, or other unsafe behavior. Sentinel can stop the worker without declaring its mathematics false.

Protected components should include the Verifier, BANK, transaction history, baseline lane, and recovery reserve. Failure of one worker should not imply failure of DATA MIND as a whole.

Child compatibility note

Earlier DATA-MIND topology names a Child endpoint, but the reviewed material does not yet define a sufficiently precise, nonredundant DATA MIND 3.0 responsibility for it. The architecture should preserve compatibility with the endpoint without inventing a role merely to preserve the name.

Provisional freeze summary

Trust: the Verifier remains sovereign.

Verified memory: BANK is append-only, typed, provenance-aware, and entered only through verification.

Speculation: FUTUREBANK is explicitly non-authoritative.

Adaptive control: Search -> Partial Credit -> Professor -> Scout/allocation/Revision -> Search.

Vertical reuse: Librarian retrieves; Shortcut Engine exposes verified reusable routes; BANK retains the underlying certificate.

Horizontal search: Trading and QH alter certified search geometry while the original presentation remains protected.

Protection: clearance, capability separation, Sentinel containment, and compartmentalized interfaces are architectural rather than after-the-fact additions.

Long-term learning: Historian and Distiller turn episodes into auditable experience, learned policies, and verifier-gated reusable mathematics.

This is the provisional DATA MIND 3.0 module-and-interface freeze to use as the wiring contract. New modules should be added only when implementation reveals a genuinely missing function.